RESPONSIVE WEB DESIGN: ARCHITECTURE & IMPLEMENTATION

Engineering Fluid User Interfaces via HTML5, CSS3, & Bootstrap 5

HTML CSS BS5

Student: CE 2nd Year | Department of Computer Engineering | Course: Web Development Frameworks | Academic Year: 2025–2026

CORE FOUNDATIONS

Core Pillars of RWD

Responsive RWD
RWD suhorts and responsive RWD

Vewport Potection
Controls wertherstands and semantic targs

HTML5 Bracters
Engineering wene/use/g interoers and mazome User optiions
Responsive, ihrizations and collection

Viewport & HTML5 Mechanics

```
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<meta name="viewport"
content="width=height-heights"
content="apexible-11" content="rous"
applicati.name="if-10">
```

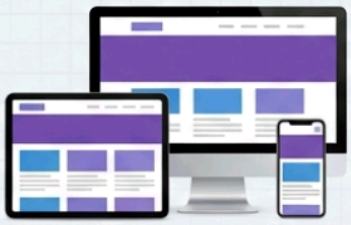
Custom badges for semantic

<header> <nav> <header>
<nav> <header> <star>
<tag> ...etc.

ARCHITECTURE & IMPLEMENTATION

CSS3 vs. Bootstrap 5

	CSS3	Bootstrap 5
Course	Comarative	Comparitive
Design	User comentration	Niv corcentration
Language	Engılish	Negative
Comrativity	Mesɔn›model	Software



Breakpoint Distribution

xs
md
lg

Mobile-First Logic

Responsive CSS **@responsive**] media queries

```
@media media queries (xs) {
  widia-height: 20px-81px;
}
@media media queries (% 80px) {
  media-height: 20px-85px;
}
```

BOOTSTRAP 5 & PERFORMANCE

Container Hierarchy

Container
Container

```
<div class="row">
  <div class="row">
    <div class="rontainers">
      <div class="rod-block"></div>
    </div>
  </div>
</div>
```

Responsive Utility Micro-Patterns

Responsive **Utility Micro-Patterns**

```
.d-none .d-md-block Resterns
.d-none .d-md-block Contantion
.d-none .d-md-block Momenlization
```

Performance Optimization

- Deromise **the Optimization** for mimate expandeed or instally
- **Performance Performance**: bost on monitoring responsive wem cally